

CONCEPT MAPPING ON OSI MODEL

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Date: 14-07-2026

Introduction

A concept map is a graphical representation of concepts and their relationships. In networking, concept maps help visualize how OSI layers interact to provide reliable, secure, and efficient communication. This assignment analyzes various concept maps related to the OSI model and networking processes.

Question 1

Missing Layers in the OSI Model

Given Concept Map

Application



Presentation



Session



Transport



Network



Data Link



Physical

Missing Layers

1. Session Layer
2. Data Link Layer

Analysis

Session Layer

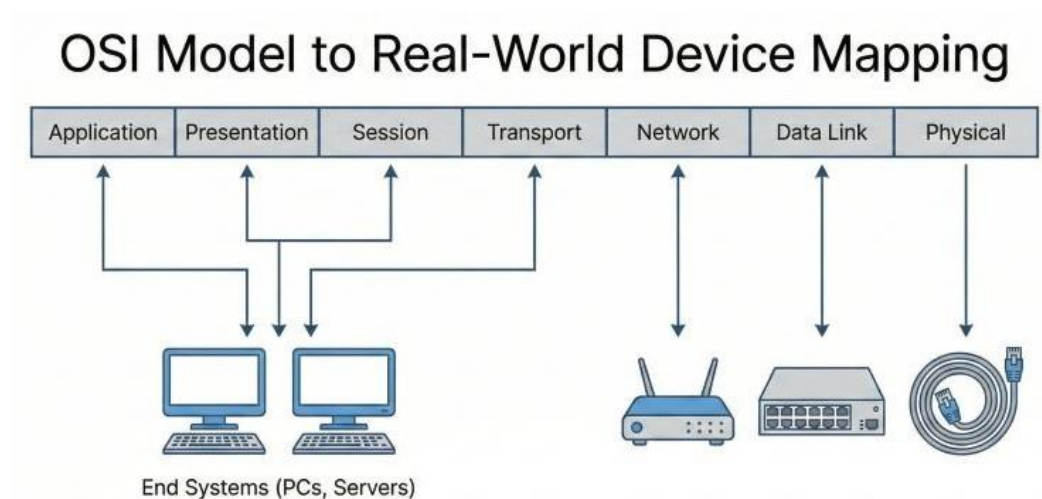
- Establishes, manages, and terminates communication sessions.
- Provides synchronization checkpoints.
- Enables recovery from communication failures.

Data Link Layer

- Converts packets into frames.
- Performs error detection using CRC.
- Controls access to the physical medium.

Importance for Reliable Communication

The Session Layer maintains communication continuity, while the Data Link Layer ensures error-free frame transmission. Together they improve reliability and efficiency.



Question 2

Data Units and Corresponding OSI Layers

Concept Map

Application Data



Transport Layer



Segment



Network Layer



Packet



Data Link Layer



Frame



Physical Layer



Bits

Data Unit	OSI Layer
Segment	Transport Layer
Packet	Network Layer
Frame	Data Link Layer

Data Unit	OSI Layer
Bits	Physical Layer

How Transformation Supports Transmission

- Transport Layer divides data into segments.
- Network Layer adds logical addressing and creates packets.
- Data Link Layer creates frames and adds error-checking information.
- Physical Layer converts frames into binary bits for transmission.

This process is known as **Encapsulation**.

Conclusion

Transformation of data units allows data to move efficiently and reliably through the network.

Question 3

MAC Address and IP Address Relationship

Concept Map

Data Link Layer



MAC Address



Local Delivery

Network Layer



IP Address



Global Delivery

Analysis

MAC Address

- Physical address of a device.
- Used within the local network.
- Ensures delivery between devices on the same LAN.

IP Address

- Logical address assigned to devices.
- Used for communication across multiple networks.
- Enables routing over the Internet.

Importance

Both addresses work together:

- IP address identifies the destination network.
- MAC address identifies the destination device.

Conclusion

The combination of MAC and IP addressing ensures successful data delivery from source to destination across local and global networks.

Address	Layer	PDU
	Application	
	Presentation	Data
	Session	
Port	Transport	Segment
IP	Network	Packet
MAC	Data Link	Frame
	Physical	Bits

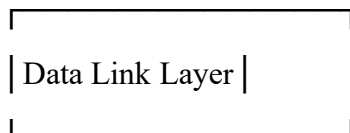
OSI model

Question 4

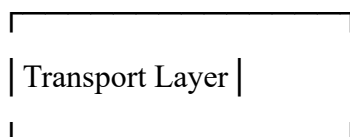
Error Control Mechanisms

Concept Map

Error Control



Error Detection



Error Correction

Analysis

Data Link Layer

- Detects transmission errors.
- Uses CRC and checksum methods.
- Identifies corrupted frames.

Transport Layer

- Performs end-to-end reliability.
- Uses acknowledgments and retransmissions.
- Corrects errors by resending lost data.

Reliability Improvement

Error detection identifies problems quickly, while error correction ensures lost or damaged data is recovered.

Conclusion

The combined operation of these layers provides highly reliable communication.

Question 5

Concept Map for Web Browsing

Concept Map

Web Browser



Application Layer (HTTP/HTTPS)



Presentation Layer (Encryption)



Session Layer (Connection Management)



Transport Layer (TCP)



Network Layer (IP Routing)



Data Link Layer (Frames)



Physical Layer (Bits)



Internet

Analysis

When a user opens a website:

1. Application Layer sends an HTTP request.
2. Presentation Layer encrypts data using SSL/TLS.
3. Session Layer maintains the connection.
4. Transport Layer ensures reliable delivery using TCP.
5. Network Layer routes packets.
6. Data Link Layer creates frames.
7. Physical Layer transmits bits through cables or wireless signals.

Failed Layer	Effect
Application	Website cannot load
Presentation	Data cannot be encrypted/decrypted
Session	Connection interruptions
Transport	Packet loss and retransmissions
Network	Routing failure
Data Link	Frame transmission errors
Physical	No network connectivity

Conclusion

All OSI layers depend on one another; failure of one layer affects the entire communication process.

Question 6

Network Troubleshooting Using Concept Mapping

Given Concept Map

Physical Layer ✓



Data Link Layer ✓



Network Layer ✗



Transport Layer ✗



Application Layer ✗

Analysis

The concept map indicates:

- Physical Layer is functioning correctly.
- Data Link Layer is functioning correctly.

- Network Layer has failed.

Because the Network Layer is not working:

- IP routing cannot occur.
- Packets cannot reach the destination.
- Transport Layer cannot establish communication.
- Applications cannot exchange data.

Possible Causes

- Incorrect IP address
- Routing table errors
- Router failure
- Network configuration issues

How Concept Mapping Helps

Concept maps show dependencies between layers. By identifying the first failed layer, network administrators can quickly locate and resolve problems.

Conclusion

Since the Network Layer fails, all upper layers also fail. Concept mapping simplifies troubleshooting and reduces diagnostic time.

Overall Conclusion

The OSI model provides a structured framework for network communication. Concept mapping helps visualize interactions among layers, addresses, data units, error-control mechanisms, and troubleshooting procedures. Understanding these relationships improves network design, maintenance, reliability, and performance.

References

1. Behrouz A. Forouzan – Data Communications and Networking.
2. Andrew S. Tanenbaum – Computer Networks.
3. IEEE Networking Standards.
4. Computer Networks Lecture Notes.